

9.12 Given that:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k p(\mathbf{x}|k)$$

$$\mathbb{E}[\mathbf{x}|k] = \boldsymbol{\mu}_k$$

$$\text{cov}[\mathbf{x}|k] = \boldsymbol{\Sigma}_k$$

The mean of the mixture distribution is given by:

$$\begin{aligned} \mathbb{E}[\mathbf{x}] &= \int p(\mathbf{x}) \mathbf{x} d\mathbf{x} \\ &= \int \sum_{k=1}^K \pi_k p(\mathbf{x}|k) \mathbf{x} d\mathbf{x} \\ &= \sum_{k=1}^K \pi_k \int p(\mathbf{x}|k) \mathbf{x} d\mathbf{x} \\ &= \sum_{k=1}^K \pi_k \mathbb{E}[\mathbf{x}|k] \\ &= \sum_{k=1}^K \pi_k \boldsymbol{\mu}_k \end{aligned}$$

which is the same as (9.49).

Using (1.42), the covariance of the mixture distribution is given by:

$$\text{cov}[\mathbf{x}] = \mathbb{E}[\mathbf{x}\mathbf{x}^T] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}]^T$$

Here,

$$\begin{aligned} \mathbb{E}[\mathbf{x}\mathbf{x}^T] &= \int \mathbf{x}\mathbf{x}^T p(\mathbf{x}) d\mathbf{x} \\ &= \int \mathbf{x}\mathbf{x}^T \sum_{k=1}^K \pi_k p(\mathbf{x}|k) d\mathbf{x} \\ &= \sum_{k=1}^K \pi_k \int \mathbf{x}\mathbf{x}^T p(\mathbf{x}|k) d\mathbf{x} \end{aligned}$$

Using $cov[\mathbf{x}|k] = \mathbb{E}[\mathbf{x}\mathbf{x}^T|k] - \mathbb{E}[\mathbf{x}|k]\mathbb{E}[\mathbf{x}|k]^T$, we have:

$$\begin{aligned} &= \sum_{k=1}^K \pi_k \left(cov[\mathbf{x}|k] + \mathbb{E}[\mathbf{x}|k]\mathbb{E}[\mathbf{x}|k]^T \right) \\ &= \sum_{k=1}^K \pi_k \left(\Sigma_k + \boldsymbol{\mu}_k \boldsymbol{\mu}_k^T \right) \end{aligned}$$

Substituting this result back into the expression for covariance of $\mathbf{x}$, we get:

$$cov[\mathbf{x}] = \left(\sum_{k=1}^K \pi_k \left(\Sigma_k + \boldsymbol{\mu}_k \boldsymbol{\mu}_k^T \right) \right) - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}]^T$$

which is the same as (9.50).

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